proper governance structure for service creation and utilisation, alert and event handling, and integration with native and third-party cost management tools and billing APIs. Tools like CloudCheckr and Apptio Cloudability fall in this category.

CloudOps driven model: This is a mid-range maturity model to provide cloud service usage optimisation or consumption gating through infrastructure optimisation like stopping idle instances, and moving compute based instances to optimised compute type to save cost. It uses spot and reserved instances for high-volume and long-term services like data lakes. It also uses IaaC templates — for instance, for VM and storage handling. Tools like Matilda, BMC TrueSight or Lightwing fall in this category.

AIOps driven model: This is the high maturity model and long-term solution to enable predictive optimisation based on usage patterns and historical data. It has an ML driven approach to create models for predictive learning and suggesting areas of improvement based on these patterns. It also creates solutions for cost optimisation for non-production environments and production environments separately for better cost benefit with low risk. Tools like Cloudhealth and Densify fall in this category.

During the FinOps life cycle, we need to design the solution to get the cost metrics, which are categorised as business metrics, and technical or usage metrics. Business metrics are a set of parameters that can help to understand the high-level view of cloud cost expense including capex and opex. Here, capex is a combination of one-time sunk cost, integration cost and locked-in cost. Capex is important to calculate the RoI in cloud investment, and both capex and opex are important to understand the total cost to the organisation.

Technically, the FinOps life cycle is handled in two stages.

Tagging and resource management: The cloud service provider gives meta data tagging or labelling for any cloud resource to derive accountability of resources, which can be used for filtering and grouping of services based on this data like environment/stage, cost centre, importance of services, application, and more.

Metrics management and usage analysis: In this stage, data analytics is done on patterns of usage over a period of time in order to find the cost optimisation patterns. This includes proactively managing resource utilisation, using historical data for cost optimisation, and integration with cost advisory tools like CloudCheckr.

For cloud cost management, FinOps helps to derive a framework to understand the cost of services utilised, track resource usage and optimise the cloud spend. This starts by onboarding a value stream of applications to the FinOps life cycle, followed by creating resource tagging and tag policies, rule-based event alerts for threshold use of the CPU, memory and storage services, and finally, preparing reporting dashboards for management or technical usage views.

For reporting dashboards, native dashboards using GCP Data Studio can be employed. The PowerBI tool can be used for easy integration with collaboration platforms like MS Teams or Atlassian Confluence. There is no need then for security activities like roles and permissions.

Cloud governance and cost economics

In the financial management of the cloud, two common heads of accounts are cost management and cost transparency. The FinOps framework implemented as part of the cloud adoption strategy addresses both of these.

There are many best practices defined for cloud cost management and cost transparency, and adopting them as part of the FinOps framework enables better cost management. The top seven ways to optimise cloud usage and control costs are listed below.

- ***Shut down unused resources/ instances:*** Though elasticity in

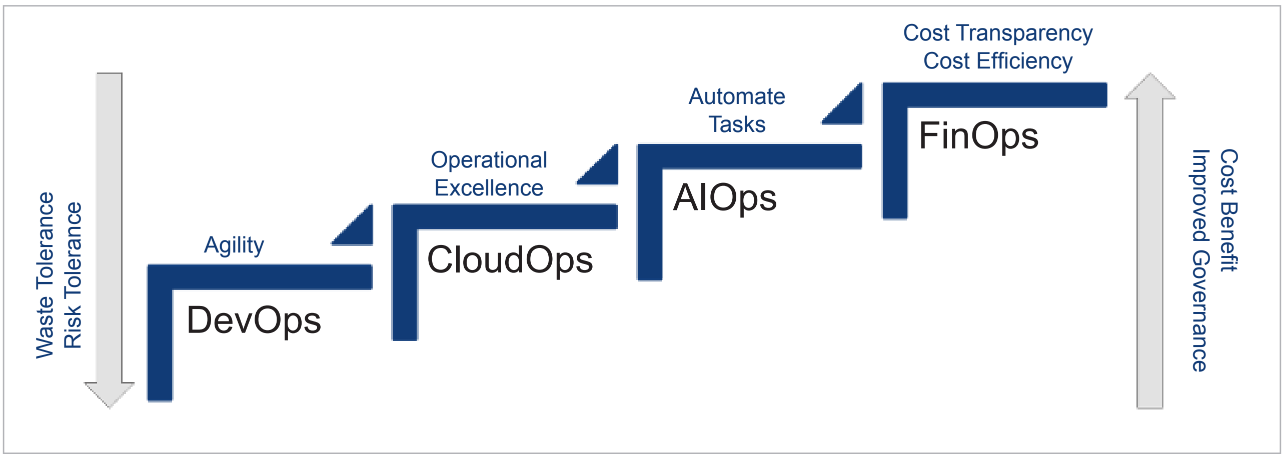

Figure 3: Cloud operations and cost economics